Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/08/2010
PI Name:	Hermanson, James C Ph.D.		
Project Title:	Stability, Cellular Structure and Heat transfer of Evaporating Films in Normal and Reduced Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
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Comments:			
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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COI Name (Institution):	Allen, Jeffrey (Michigan Technological University)		
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	This program studies fundamental film evaporation phenomena under normal- and reduced-gravity conditions. Experimental tests, combined with detailed stability analysis, are employed to achieve an improved understanding of the fluid physics responsible for film instability, changes in the film structure, and the corresponding implications for heat transfer, and how those are impacted by the removal of buoyancy in reduced gravity. The mass loss and vapor recoil due to evaporation and potential changes in surface temperature owing to phase change are expected to play a de-stabilizing role for films in reduced gravity, where surface-tension driven convection can become important. The impact of these mechanisms on the stability and structure for both buoyant and non-buoyant films are examined by the systematic variation of the wall temperature and the nature of the heat input (wall surface heating versus system depressurization). The effects of non-condensable gases on film evaporation will also be examined. Both steady-state and transient film		

Task Description:	evaporation phenomena are studied. A comprehensive suite of instrumentation is used for the systematic examination of the fluid physics described above. These diagnostics include temperature and heat transfer measurements as well as ultrasonic measurement of film thickness. Optical techniques are employed to determine the features of the resulting instabilities, variations in surface temperature, and the evolution of the cellular structure within the evaporating films. In parallel with the experimental program, a stability analysis is performed to model the film evaporation and the development of interfacial disturbances. The analysis will be used both to aid in the interpretation of the ground-based results, as well as to shed important insight into the evaporating film behavior to be expected under long-term, microgravity conditions. The proposed ground-based investigation will conclude with the recommendation of a preliminary microgravity flight experiment design to NASA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The high heat flux associated with phase change is an important consideration in spacecraft thermal control applications. Devices such as capillary pumped loops and heat pipes depend critically on evaporation and the successful utilization of these devices in space applications require a detailed understanding of evaporation phenomena under reduced gravity conditions. Other important technologies related to space exploration requiring a more in-depth understanding of evaporation include humidity control of cabin environment, water recovery from waste sources, oxygen production, water management in power systems such as fuel cells, and in-situ resource utilization for fuel production. Understanding evaporation phenomena is critical to the development and optimization of these space technologies that enable human exploration of space. In addition to space applications, evaporation is important to many terrestrial technologies from drying applications in the pulp and paper industry to coating deposition on fiber optics. Many of these technologies, while reliable, are not optimized for lack of a thorough understanding of evaporation phenomena. International competition is driving a need beyond consistent, reliable operation of manufacturing and processing technologies towards more optimized operations. Research on evaporation mechanisms may assist in this optimization. The use of ultrasound for the dynamic monitoring of film thickness and growth rate developed as part of this effort has potentially widespread use for the determination of film behavior in a variety of devices that employ two-phase flow, such as power stations and closed-loop refrigeration cycles, evaporators, and condensers.
	Laboratory Experiments Film evaporation experiments over the past year were conducted in the laboratory using a closed test cell with controlled pressure and a heated wall. Six different working fluids were utilized: n-pentane, methanol, dichloromethane, chloroform, diethyl ether, and acetone. These fluids were chosen for their wide varieties in properties, relatively high vapor pressures, good wetting abilities, relative ease of handling, and because they are readily available. Diagnostics included ultrasound film thickness measurement, surface and vapor thermocouple temperature measurement, and double-pass schlieren imaging. An image processing technique was developed to allow for the quantitative determination of the interfacial wavelengths associated with evaporating films. The technique involves median-value subtraction, followed by adaptive filtering with an adaptive 30x30 Wiener two-dimensional filter. A two dimensional FFT is then performed transferring the image to frequency space. Finally, the averages over rings of constant frequency are taken, and the frequency is inverted to yield plots of the magnitude versus wavelength. This technique has successfully extracted wavelength information for both transient and steady-state evaporating films in the laboratory. Transient film evaporation experiments were conducted by rapidly lowering the system pressure, in a fashion similar to that in the planned reduced-gravity experiments. These experiments focused on the study of the onset of interfacial disturbances, and the development and penetration of the convective structures into the bulk of the film. The transient films generally exhibit two sharp rises in heat flux. The first rise is associated with the transient superheating; the second, by a marked change in the convective structure, which typically leads to a peak heat flux condition. The wall boundary condition does not appear to be important in these films at the onset of convection. Simple, 1-D modeling was performed to help understand the magnitude of the peak heat flux, the time delay to the peak, and the onset of convective instability. This analysis allowed for estimation of the amount of heat flux from the wall during these transient experiments, where the heat flux at the wall would be expected to be different from that at the evaporating surface. The transient films exhibited convective structures similar in many respects to those seen in evaporating films under quasi-steady conditions, after a sufficiently long elapsed time. The observed similarity between the convective structures for the transient and quasi-steady evaporating films suggests a mean of estimating the penetration depth of the convective structures into the bulk of the film for the transient evaporating films. This was done by identifying the quasi-steady film of known thickness that exhibited similar instability wavelengths to that seen for a given transient film. That thickness was then taken to be the expected “penetration thickness” of the convective structures into the bulk film at a given stage of the transient film evolution. Limited tests with non-evaporating films heated from below were conducted to verify parts of the experimental setup and to provide baseline information on convective structure against which the evaporating film results might be compared. High temperature silicone oil (100 to 150 cSt) was introduced onto the copper surface and its thickness was measured using ultrasound. The liquid was heated from below using circulating warm water and cooled from above by the ambient air. The convective environment at the free surface allowed for only a limited temperature difference across the liquid layer. This precluded the development of the more complicated structures (spiral structures, rolls, etc.) seen in the evaporating films. Preliminary laboratory experiments were conducted with liquid films evaporating into ambient air, where the presence of non-condensible gases can be expected to have a significant effect on the evaporation process. In these studies, the cover window was removed, the test cell wall was shortened, and liquids were poured onto the copper substrate and allowed to evaporate directly into the atmosphere. The evaporation rate in this case appears to be limited by diffusion of

Task Progress:	the working fluid vapor into the environment. Aircraft experiment development The aircraft test right for the study of evaporating thin films in reduced gravity is currently in the mid stages of its development. The central working components are the test chamber, in which the film evaporation occurs, and an expansion chamber to allow for evaporation in a constant-pressure condition. The glass test chamber allows for optical access for visualization of the evaporating film. Heat input to the film is provided by thermoelectric heaters/coolers mounted below the flat, copper surface. A second thermoelectric cooler is mounted at the side of the test chamber to allow for the condensation of evaporated fluid to minimize the volume expansion. The main frame, evaporating film test chamber, expansion chamber have been assembled, and preliminary operational tests have been conducted. The expansion chamber has been designed for better sealing and improved accessibility for more rapid assembly. The basic experiment to be conducted under reduced-gravity conditions, that is, the evaporation of a fluid by lowering the pressure, has been tested in the laboratory at the upper limit of the pressure range at which the experiment is expected to operate. This was done first with water as the working fluid, and subsequently, with acetone later. In both cases the thermoelectric coolers worked as designed. A data acquisition system was connected to the electronic valve used to regulate the pressure inside the expansion chambers. Effort has also been directed at developing a suitable bladder for the expansion chamber. The expansion of the bladder, under pressure conditions regulated in the expansion chamber, allows for the volume released by the evaporating film. A preliminary bladder, constructed from latex, works as designed. The use of materials that will provide for improved compatibility with the different test fluids under consideration is being investigated. The design for the integration of the ultrasound, thermal, and optical diagnostics into the aircraft test rig is underway. In the same fashion as the laboratory experiments, ultrasound will be used to perform dynamic measurements of the evaporating film thickness. Thermal measurements will be performed using thermocouples, and optical imaging of the evaporating films will be provided by double-pass schlieren imaging. The efficacy of all of these techniques in the investigation of evaporating films has been demonstrated by the experiments conducted in the laboratory described above. Numerical Modeling and Linear Stability Theory Linear stability theory is being utilized in this work to provide important information on the expected instability wavelengths, changes in layer thickness due to evaporation, and the expected film behavior for the different thermal and physical boundary conditions. The major areas of work have been in the investigation of different methods of solutions of the governing equations and the application of these methods to the particular case of interest. The spatiotemporal evolution equation is being used to model interfacial instabilities. A finite difference code has been developed to solve the non-linear spatiotemporal equation as an initial value problem subject to periodic or pinned boundary conditions. The numerical method is an implicit central differencing scheme that is forward differencing in time. The code is used to isolate various effects as described in nondimensional parameters such gravitational, surface tension, disjoining pressure, and evaporation. A study of the maximizing wavenumbers causing film instability under conditions of film evaporation is performed using linear stability theory. Rupture of an isothermal, thin film under the effect of destabilizing van der Waals force and the stabilizing effect of surface tension has been demonstrated. In a similar fashion to the isothermal film, the numerical code has been successfully used to evolve a film under the influence of gravity and surface tension alone. Thick films with evaporation, gravitational acceleration and surface tension are modeled by solving the spatiotemporal evolution equation as an initial value problem. The evaporating thick film has also been modeled using pinned boundary conditions instead of periodic boundary conditions to establish the influence of pinned boundary conditions on maximizing wavenumber and the effect of wavelength versus domain size.
Bibliography Type:	Description: (Last Updated: 09/03/2025)
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